

1- El movimiento relativista de una partícula en un potencial estático $V(\vec{r})$ puede ser obtenido a partir del Lagrangiano

$$\mathcal{L} = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - V(\vec{r})$$

(a) Escriba las ecuaciones de Euler-Lagrange y Verifique afirmación anterior

(b) Encuentre movimiento Canónico $\vec{p}$ y muestre que el Hamiltoniano relativista viene dado por

$$H = \sqrt{m^2 c^4 + p^2 c^2} + V(\vec{r})$$

¿Es H una constante de mov.?

(c) Asumir que V es esféricamente simétrico. Muestre que $\vec{r} \times \vec{p}$
De esta forma, reducida a la forma

$$H = c \sqrt{m^2 c^2 + p_r^2 + \frac{p_\phi^2}{r^2}} + V(\vec{r})$$

a) $\mathcal{L} = -mc^2 \left(1 - \frac{v^2}{c^2}\right)^{1/2} - V(\vec{r})$; si $v^2 \ll c^2 \Rightarrow \left(1 - \frac{v^2}{c^2}\right)^{1/2} \approx 1 - \frac{v^2}{2c^2}$; $\frac{v^2}{c^2} \ll 1$

$$\therefore \mathcal{L} = -mc^2 \left[1 - \frac{v^2}{2c^2}\right] - V(\vec{r}) = \underbrace{-mc^2}_{\text{cte}} + \frac{mv^2}{2} - V(\vec{r})$$

cte $\rightarrow$ no afecta ec. de mov.

$$\therefore \boxed{\mathcal{L} = \frac{1}{2}mv^2 - V(\vec{r})}$$
 Lagrangiano clásico

Ecuaciones de movimiento: $\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \vec{v}} \right) - \frac{\partial \mathcal{L}}{\partial \vec{r}} = 0$; $\vec{v} \cdot \vec{v} = v^2$

$$\frac{\partial \mathcal{L}}{\partial \vec{v}} = \frac{\partial}{\partial \vec{v}} \left[-mc^2 \left(1 - \frac{\vec{v} \cdot \vec{v}}{c^2}\right)^{1/2} - V(\vec{r}) \right] = -\frac{mc^2}{2} \frac{(-2 \frac{\vec{v}}{c^2})}{\left(1 - \frac{\vec{v} \cdot \vec{v}}{c^2}\right)^{1/2}} = \frac{m\vec{v}}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}}$$

$$\frac{\partial \mathcal{L}}{\partial \vec{r}} = -\vec{\nabla} V(\vec{r}) \quad ; \quad \vec{p} = \frac{\partial \mathcal{L}}{\partial \vec{v}}$$

$$\therefore \boxed{\frac{d}{dt} \left[\frac{m\vec{v}}{\left(1 - \frac{v^2}{c^2}\right)^{1/2}} \right] + \vec{\nabla} V(\vec{r}) = 0}$$

Ecuación Euler-Lagrange

$$b) \vec{p} = \partial_{\vec{v}} \mathcal{L}, \text{ momento canónico}$$

$$\vec{p} = \frac{m \vec{v}}{(1 - \frac{v^2}{c^2})^{1/2}}$$

$$\text{Hamiltoniano: } H = \vec{p} \cdot \vec{v} - \mathcal{L}$$

$$H = \frac{m \vec{v} \cdot \vec{v}}{(1 - \frac{v^2}{c^2})^{1/2}} + mc^2 \left(1 - \frac{v^2}{c^2}\right)^{1/2} + V(\vec{r})$$

$$= \frac{mv^2 + mc^2 - mv^2}{(1 - \frac{v^2}{c^2})^{1/2}} + V(\vec{r}) \Rightarrow H = \frac{mc^2}{(1 - \frac{v^2}{c^2})^{1/2}} + V(\vec{r})$$

$$\text{Truquito: } \frac{mc^2}{(1 - \frac{v^2}{c^2})^{1/2}} = \sqrt{\frac{m^2 c^4}{(1 - \frac{v^2}{c^2})} - p^2 c^2 + p^2 c^2} = \sqrt{\frac{m^2 c^4}{(1 - \frac{v^2}{c^2})} - \frac{m^2 v^2 c^2}{(1 - \frac{v^2}{c^2})} + p^2 c^2}$$

$$= \sqrt{m^2 c^4 \left(\frac{1}{(1 - \frac{v^2}{c^2})} - \frac{v^2/c^2}{(1 - \frac{v^2}{c^2})} \right) + p^2 c^2} = \sqrt{m^2 c^4 + p^2 c^2} + V(\vec{r})$$

$$\Rightarrow H = \sqrt{m^2 c^4 + p^2 c^2} + V(\vec{r}) ; \text{ } \mathcal{L} \text{ no depende del tiempo, entonces } H = E \text{ y por lo tanto, es una cte de mov. } \boxed{\frac{dH}{dt} = 0}$$

$$c) \vec{L} = \vec{r} \times \vec{p} ; \frac{d}{dt} (\vec{r} \times \vec{p}) = \underbrace{\dot{\vec{r}} \times \vec{p}}_0 + \underbrace{\vec{r} \times \dot{\vec{p}}}_{\vec{r} \times \nabla V(\vec{r}) = 0} ; \vec{p} = m \vec{r} \dot{\vec{v}} \parallel \vec{v}, \dot{\vec{p}} = -\nabla V(\vec{r})$$

$$\frac{d\vec{L}}{dt} = 0 \Rightarrow \boxed{\vec{L} = \text{cte mov.}}$$

$$p^2 = p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2 \sin^2 \theta} \xrightarrow[\theta = \pi/2 \rightarrow p_\theta = 0]{\text{mov. en un plano } V(r) \hat{r}} p^2 = p_r^2 + \frac{p_\phi^2}{r^2}$$

$$\therefore H = c \sqrt{m^2 c^2 + p^2} + V(r) = c \sqrt{m^2 c^2 + p_r^2 + \frac{p_\phi^2}{r^2}} + V(r)$$

$$\Rightarrow \boxed{H = c \sqrt{m^2 c^2 + p_r^2 + \frac{p_\phi^2}{r^2}} + V(\vec{r})}$$

$$2- T = \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2)(q_1^2 + q_2^2), \quad V = (q_1^2 + q_2^2)^{-1} = \frac{1}{q_1^2 + q_2^2}$$

$\{q_1, q_2\}$ coord. gen.

(a) ¿Cuál es la ec. de Hamilton-Jacobi para este sistema?

(b) Encuentre la función principal de Hamilton

(c) Deducir el mov. dinámico del sistema (no se necesita ninguna integral de mov.)

$$(a) H = \sum_j p_j \dot{q}_j - L; \quad L = T - V = \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2)(q_1^2 + q_2^2) - \frac{1}{q_1^2 + q_2^2}$$

$$p_1 = \partial_{\dot{q}_1} L = \dot{q}_1(q_1^2 + q_2^2), \quad H = p_1 \dot{q}_1 + p_2 \dot{q}_2 - \frac{(\dot{q}_1^2 + \dot{q}_2^2)(q_1^2 + q_2^2)}{2} + \frac{1}{q_1^2 + q_2^2}$$

$$p_2 = \partial_{\dot{q}_2} L = \dot{q}_2(q_1^2 + q_2^2), \quad \dot{q}_1 = \frac{p_1}{(q_1^2 + q_2^2)} (*), \quad \dot{q}_2 = \frac{p_2}{(q_1^2 + q_2^2)} (\Delta)$$

$$\therefore H = p_1 \dot{q}_1 + p_2 \dot{q}_2 - \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2)(q_1^2 + q_2^2) + \frac{1}{q_1^2 + q_2^2}$$

$$= \underbrace{\dot{q}_1}_{(*)} \left(\underbrace{p_1 - \frac{1}{2}\dot{q}_1(q_1^2 + q_2^2)}_{(*) \rightarrow \frac{1}{2}p_1} \right) + \underbrace{\dot{q}_2}_{(\Delta)} \left(\underbrace{p_2 - \frac{1}{2}\dot{q}_2(q_1^2 + q_2^2)}_{(\Delta) \rightarrow \frac{1}{2}p_2} \right) + \frac{1}{q_1^2 + q_2^2}$$

$$\Rightarrow H = \frac{p_1^2}{2(q_1^2 + q_2^2)} + \frac{p_2^2}{2(q_1^2 + q_2^2)} + \frac{1}{q_1^2 + q_2^2}$$

Hamiltoniano

$S \rightarrow$ función principal de Hamilton

cte de mov.

$$* p_1 = \partial_{\dot{q}_1} S, \quad p_2 = \partial_{\dot{q}_2} S, \quad H = -\partial_t S; \quad H(\partial_{q_1} S, \dots, \partial_{q_n} S; q_1, \dots, q_n; t) + \partial_t S = 0$$

$$\Rightarrow -\partial_t S = \frac{1}{2(q_1^2 + q_2^2)} \left[(\partial_{q_1} S)^2 + (\partial_{q_2} S)^2 \right] + \frac{1}{q_1^2 + q_2^2} \quad \text{Ecuación Hamilton-Jacobi}$$

$$\Rightarrow \left[\frac{1}{2(q_1^2 + q_2^2)} \left[(\partial_{q_1} S)^2 + (\partial_{q_2} S)^2 \right] + \frac{1}{q_1^2 + q_2^2} + \partial_t S \right] = 0 \quad \text{Hamilton Jacobi}$$

$$\text{Ansatz: } S(\{q\}, \alpha, t) = \underbrace{W(\{q\}, \alpha)}_{\text{función característica de Hamilton}} - \alpha t; \quad H(\partial_{q_1} W, \partial_{q_2} W, q_1, q_2) = \alpha = E$$

función característica de Hamilton

de integración

$$W(q_1, \dots, q_n) = W_1(q_1) + W_2(q_2) + \dots + W_n(q_n)$$

cond. separabilidad

$$\therefore [(\partial_{q_1} W)^2 + (\partial_{q_2} W)^2] = 2E(q_1^2 + q_2^2) - 2$$

b) Sistema circularmente simétrico, por la forma de T y V

→ sdo dep. $r^2 = q_1^2 + q_2^2$; $\begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} q_1' \\ q_2' \end{pmatrix}$

$\Rightarrow \underbrace{q_1^2 + q_2^2 = q_1'^2 + q_2'^2} \quad \left\{ \|\vec{q}\|^2 = \vec{q}^T \vec{q}' = (\mathbf{R} \vec{q}')^T (\mathbf{R} \vec{q}') = \vec{q}'^T \vec{q}' = q_1'^2 + q_2'^2 \right\}$

Invarianza bajo rotaciones $\rightarrow$ Noether: Simetría rotacional $\rightarrow$ Conservación momentum

$(q_1, q_2) \rightarrow (r, \theta)$

$\therefore$ Si $q_1 = r \cos \theta$, $q_2 = r \sin \theta$; $r = \sqrt{q_1^2 + q_2^2}$, $\theta = \text{Arctan}\left(\frac{q_2}{q_1}\right)$

$S(r, \theta) : \partial_{q_1} S = \partial_{q_1} r \underbrace{\partial_r S}_{S_r} + \partial_{q_1} \theta \underbrace{\partial_\theta S}_{S_\theta}$; $\partial_{q_2} S = \partial_{q_2} r \underbrace{\partial_r S}_{S_r} + \partial_{q_2} \theta \underbrace{\partial_\theta S}_{S_\theta}$

$\partial_{q_1} r = \frac{q_1}{r} = \cos \theta$

$\partial_{q_1} \theta = \frac{1}{1+u^2} \frac{\partial u}{\partial q_1} = \frac{1}{1+(\frac{q_2}{q_1})^2} \left(-\frac{q_2}{q_1^2}\right)$

$\partial_{q_2} r = \frac{q_2}{r} = \sin \theta$

$= \frac{q_1^2}{r^2} \left(-\frac{q_2}{q_1^2}\right) = -\frac{q_2}{r^2} = -\frac{\sin \theta}{r}$

$\partial_{q_2} \theta = \frac{q_1}{r^2} \left(\frac{1}{q_1}\right) = \frac{q_1}{r^2} = \frac{\cos \theta}{r}$

$\therefore \partial_{q_1} S = \cos \theta S_r - \frac{\sin \theta}{r} S_\theta$; $\partial_{q_2} S = \sin \theta S_r + \frac{\cos \theta}{r} S_\theta$

$\Rightarrow (\partial_{q_1} S)^2 + (\partial_{q_2} S)^2 = \left(\cos \theta S_r - \frac{\sin \theta}{r} S_\theta\right)^2 + \left(\sin \theta S_r + \frac{\cos \theta}{r} S_\theta\right)^2$

$= \left(\cos^2 \theta S_r^2 - \frac{2 \sin \theta \cos \theta}{r} S_r S_\theta + \frac{\sin^2 \theta}{r^2} S_\theta^2\right) + \left(\sin^2 \theta S_r^2 + \frac{2 \sin \theta \cos \theta}{r} S_r S_\theta + \frac{\cos^2 \theta}{r^2} S_\theta^2\right)$

$= S_r^2 + \frac{1}{r^2} S_\theta^2 = (\partial_r S)^2 + \frac{1}{r^2} (\partial_\theta S)^2$

$\Rightarrow (\partial_r S)^2 + \frac{1}{r^2} (\partial_\theta S)^2 = 2E r^2 - 2$; Separación de Variables: $S(r, \theta) = S_r(r) + \Theta(\theta)$

$\Rightarrow (S_r')^2 + \frac{1}{r^2} (\Theta')^2 = 2E r^2 - 2$; $\Theta' = l \Rightarrow \Theta(\theta) = l\theta$ $\left\{ \begin{array}{l} (\Theta')^2 = l^2 \\ (\Theta')^2 = l^2 \end{array} \right.$

$\Rightarrow (S_r')^2 = 2E r^2 - 2 - \frac{l^2}{r^2} \Rightarrow S_r(r) = \int dr \sqrt{2E r^2 - 2 - \frac{l^2}{r^2}}$

$\therefore S(r, \theta) = l\theta + \int dr \sqrt{2E r^2 - 2 - \frac{l^2}{r^2}}$; $\Theta'(\theta) = \partial_\theta S = p_\theta$ Momento canónico conjugado

función principal de Hamilton

$p_\theta = l$ cte mov. asociada a la simetría rotacional (Noether): invarianza bajo rotaciones

c) Mov. dinámico

$$p_\theta = \partial_\theta S = l = \text{cte} \Rightarrow \text{Conservación momento } \neq$$

$$p_r = \partial_r S = \sqrt{2Er^2 - 2 - \frac{l^2}{r^2}}, \text{ momento radial}$$

Ecuaciones de Hamilton,

$$\dot{r} = \partial_{p_r} H = \frac{p_r}{r^2}$$

$$\Rightarrow \dot{r} = \frac{1}{r^2} \sqrt{2Er^2 - 2 - \frac{l^2}{r^2}}$$

$$\dot{\theta} = \partial_{p_\theta} H = \frac{p_\theta}{r^2} = \frac{l}{r^2}$$

$$\frac{d\theta}{dr} = \frac{\dot{\theta}}{\dot{r}} = \frac{l}{\sqrt{2Er^2 - 2 - \frac{l^2}{r^2}}}$$

Hamiltoniano

1- Una partícula de masa m se mueve en un campo de fuerzas cuyo potencial en coor. esféricas es

$$V(r, \theta) = - \frac{K \cos \theta}{r^2}$$

Obtener ecuaciones de mov. canónicas

$$L = T - V$$

$$T = \frac{1}{2} m v^2 = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2)$$

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) + \frac{K \cos \theta}{r^2} \quad ; \quad H = \sum_j p_j \dot{q}_j - L$$

$$\therefore H = \dot{r} p_r + \dot{\theta} p_\theta + \dot{\phi} p_\phi - \frac{1}{2} m \dot{r}^2 - \frac{1}{2} m r^2 \dot{\theta}^2 - \frac{1}{2} m r^2 \sin^2 \theta \dot{\phi}^2 - \frac{K \cos \theta}{r^2}$$
$$= \dot{r} (p_r - \frac{1}{2} m \dot{r}) + \dot{\theta} (p_\theta - \frac{1}{2} m r^2 \dot{\theta}) + \dot{\phi} (p_\phi - \frac{1}{2} m r^2 \sin^2 \theta \dot{\phi}) - \frac{K \cos \theta}{r^2}$$

Ecs. mov. Canon: $\partial_{\dot{q}_j} L = p_j$, $\dot{p}_j = -\partial_{q_j} H$, $\dot{q}_j = \partial_{p_j} H$

$$\Rightarrow p_r = \partial_{\dot{r}} L = m \dot{r} \Rightarrow \dot{r} = \frac{p_r}{m}$$

$$\Rightarrow p_\theta = \partial_{\dot{\theta}} L = m r^2 \dot{\theta} \Rightarrow \dot{\theta} = \frac{p_\theta}{m r^2}$$

$$\Rightarrow p_\phi = \partial_{\dot{\phi}} L = m r^2 \sin^2 \theta \dot{\phi} \Rightarrow \dot{\phi} = \frac{p_\phi}{m r^2 \sin^2 \theta}$$

$$\therefore H = \frac{p_r}{m} \left(p_r - \frac{1}{2} m \frac{p_r}{m} \right) + \frac{p_\theta}{m r^2} \left(p_\theta - \frac{1}{2} m r^2 \frac{p_\theta}{m r^2} \right) + \frac{p_\phi}{m r^2 \sin^2 \theta} \left(p_\phi - \frac{1}{2} m r^2 \sin^2 \theta \frac{p_\phi}{m r^2 \sin^2 \theta} \right) - \frac{K \cos \theta}{r^2}$$

$$H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2m r^2} + \frac{p_\phi^2}{2m r^2 \sin^2 \theta} - \frac{K \cos \theta}{r^2}$$

$$\therefore \dot{p}_r = -\partial_r H = \frac{p_\theta^2}{m r^3} + \frac{p_\phi^2}{m r^3 \sin^2 \theta} - \frac{2K \cos \theta}{r^3}$$

$$\dot{p}_\phi = -\partial_\phi H = 0 \quad , \quad \dot{p}_\theta = \frac{p_\phi^2 \cos \theta}{m r^2 \sin^3 \theta} - \frac{K \sin \theta}{r^2}$$

$$\dot{r} = \partial_{p_r} H = \frac{p_r}{m} \quad ; \quad \dot{\theta} = \partial_{p_\theta} H = \frac{p_\theta}{m r^2} \quad ; \quad \dot{\phi} = \frac{p_\phi}{m r^2 \sin^2 \theta}$$

Galactic Dynamics, J.S. Binney: 3.5.3 Action

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Las coordenadas esferoidales-oblatas (u, v, ϕ) están relacionadas a las coordenadas cilíndricas (R, z, ϕ) por

$$R = \Delta \cosh u \cos v, \quad z = \Delta \sinh u \sin v \quad \cosh^2 u - \sinh^2 u = 1$$

(a) demostrar $p_u = m \Delta^2 (\cosh^2 u - \cos^2 v) \dot{u}$ $\hookrightarrow \sinh^2 u = \cosh^2 u - 1$

$$p_v = m \Delta^2 (\cosh^2 u - \cos^2 v) \dot{v}$$

$$p_\phi = m \Delta^2 \cosh^2 u \cos^2 v \dot{\phi}$$

$$\star \mathcal{L}(\vec{x}, \vec{\dot{x}}, t) = \frac{1}{2} m \vec{\dot{x}}^2 - V(\vec{x}, t); \quad \vec{p} = \left(\frac{\partial \mathcal{L}}{\partial \vec{\dot{x}}} \right)_{\vec{x}, t}$$

$$H(\vec{q}, \vec{p}, t) \equiv \vec{p} \cdot \vec{\dot{q}} - \mathcal{L}(\vec{q}, \vec{\dot{q}}, t)$$

$$\mathcal{L}(R, z, \phi) = \frac{1}{2} m \{ \dot{R}^2 + R^2 \dot{\phi}^2 + \dot{z}^2 \} - \Phi$$

$$R = \Delta \cosh u \cos v \rightarrow \dot{R} = \dot{u} \Delta \sinh u \cos v - \dot{v} \Delta \cosh u \sin v$$

$$z = \Delta \sinh u \sin v \rightarrow \dot{z} = \dot{u} \Delta \cosh u \sin v + \dot{v} \Delta \sinh u \cos v$$

$$\begin{aligned} \dot{R}^2 + \dot{z}^2 &= (\dot{u} \Delta \sinh u \cos v - \dot{v} \Delta \cosh u \sin v)^2 + (\dot{u} \Delta \cosh u \sin v + \dot{v} \Delta \sinh u \cos v)^2 \\ &= \dot{u}^2 \Delta^2 \sinh^2 u \cos^2 v - 2 \dot{u} \dot{v} \Delta^2 \sinh u \cosh u \cos v \sin v + \dot{v}^2 \Delta^2 \cosh^2 u \sin^2 v + \\ &\quad \dot{u}^2 \Delta^2 \cosh^2 u \sin^2 v + 2 \dot{u} \dot{v} \Delta^2 \cosh u \sinh u \sin v \cos v + \dot{v}^2 \Delta^2 \sinh^2 u \cos^2 v \\ &= \Delta^2 \dot{u}^2 \{ \cosh^2 u \sin^2 v + \sinh^2 u \cos^2 v \} + \Delta^2 \dot{v}^2 \{ \cosh^2 u \sin^2 v + \sinh^2 u \cos^2 v \} \\ &= \Delta^2 \dot{u}^2 \{ (\cosh^2 u - 1)(1 - \sin^2 v) + \cosh^2 u \sin^2 v \} + \Delta^2 \dot{v}^2 \{ \cosh^2 u (1 - \cos^2 v) + \sinh^2 u \cos^2 v \} \\ &= \Delta^2 \{ \dot{u}^2 + \dot{v}^2 \} (\cosh^2 u - \cos^2 v) \end{aligned}$$

$$\therefore \mathcal{L} = \frac{1}{2} m \Delta^2 \{ (\cosh^2 u - \cos^2 v) (\dot{u}^2 + \dot{v}^2) + \cosh^2 u \cos^2 v \dot{\phi}^2 \} - \Phi$$

$$\vec{p} = \left(\frac{\partial \mathcal{L}}{\partial \vec{\dot{q}}} \right);$$

$$\Rightarrow u = \partial_{\dot{u}} \mathcal{L} = m \Delta^2 (\cosh^2 u - \cos^2 u) \dot{u} ; p_\phi = m \Delta^2 \cosh^2 u \cos^2 u \dot{\phi}$$

$$p_u = \partial_{\dot{u}} \mathcal{L} = m \Delta^2 (\cosh^2 u - \cos^2 u) \dot{u}$$

$$\dot{u} = \frac{p_u}{m \Delta^2 (\cosh^2 u - \cos^2 u)} ; \dot{u} = \frac{p_u}{m \Delta^2 (\cosh^2 u - \cos^2 u)} ; \dot{\phi} = \frac{p_\phi}{m \Delta^2 \cosh^2 u \cos^2 u}$$

$$H = \sum_j p_j \dot{q}_j - \mathcal{L} = p_u \dot{u} + p_u \dot{u} + p_\phi \dot{\phi} - \frac{1}{2} m \Delta^2 (\cosh^2 u - \cos^2 u) \dot{u}^2 - \frac{1}{2} m \Delta^2 (\cosh^2 u - \cos^2 u) \dot{u}^2 - \frac{1}{2} m \Delta^2 \cosh^2 u \cos^2 u \dot{\phi}^2 + \Phi$$

$$= \dot{u} (p_u - \frac{1}{2} m \Delta^2 (\cosh^2 u - \cos^2 u) \dot{u}) + \dot{u} (p_u - \frac{1}{2} m \Delta^2 (\cosh^2 u - \cos^2 u) \dot{u}) + \dot{\phi} (p_\phi - \frac{1}{2} m \Delta^2 \cosh^2 u \cos^2 u \dot{\phi}) + \Phi$$

$$= \frac{p_u}{m \Delta^2 (\cosh^2 u - \cos^2 u)} \left\{ p_u - \frac{1}{2} p_u \right\} + \frac{p_u}{m \Delta^2 (\cosh^2 u - \cos^2 u)} \left\{ p_u - \frac{1}{2} p_u \right\} + \frac{p_\phi}{m \Delta^2 \cosh^2 u \cos^2 u} \left\{ p_\phi - \frac{1}{2} p_\phi \right\} + \Phi$$

$$\Rightarrow H = \frac{p_u^2 + p_u^2}{2 m \Delta^2 (\cosh^2 u - \cos^2 u)} + \frac{p_\phi^2}{2 m \Delta^2 \cosh^2 u \cos^2 u} + \Phi$$